Dentigo — Software Requirements Specification

Version 1.4 | Secretary, Appointments and Clinic Payment Records

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AI Integration: No AI

Document Status: Working Baseline for UI Design and Implementation

1. Introduction

1.1 Purpose

This document describes the functional and non-functional requirements of Dentigo, a web-based Dental Case Quality and Treatment Management System.

It defines the system's users, main features, business rules, case workflows, data requirements, and access permissions. It serves as a reference for UI/UX design, backend development, database design, and system testing.

1.2 Project Scope

Dentigo aims to organize dental treatment cases and improve communication between dentists, dental laboratories, and radiology users.

The system allows dentists to create dental cases, attach relevant information and files, send cases to dental laboratories, review submitted designs, request revisions, and approve designs.

Radiology users can upload radiology images and reports when required. Dental laboratories can access the information and files needed to prepare dental designs. The system maintains case history and file versions to support traceability and reduce confusion.

Dentigo also provides dashboards, notifications, case search, and administrative account management.

The project includes a web interface and a backend API connected to a relational database.

1.3 Definitions

• Dental Case: A record representing a specific treatment or procedure for one patient and one tooth.

• Dentist / Doctor: A user responsible for creating cases, managing clinical information, reviewing designs, and making treatment-related decisions.

• Lab Technician: A user responsible for processing dental cases and submitting dental designs.

• Radiology User: A user responsible for uploading required radiology images and reports.

• Administrator: A user responsible for account management and system monitoring.

• Design Version: A submitted dental design saved as a distinct version.

• Case History: A chronological record of important actions and changes related to a case.

• Quality Gate: A validation step that checks whether the required information and files are available before a case is submitted to the laboratory.

• MVP: Minimum Viable Product, representing the essential features included in the first version.

1.4 Intended Audience

This document is intended for:

• UI/UX designers.

• Backend developers.

• Database designers.

• Project supervisors and stakeholders.

• Software testers.

1.5 Technical Environment

The proposed system consists of:

• A web-based user interface.

• ASP.NET Core Web API for backend services.

• A relational database.

• REST API endpoints for communication between the frontend and backend.

• Server-side authentication and authorization.

The UI design is prepared in Figma. The design itself does not mean that the corresponding functionality has already been implemented.

2. Overall Description

2.1 Product Perspective

Dentigo is designed to centralize dental case information and organize the workflow between dentists, dental laboratories, and radiology users.

Each Dental Case is associated with one patient and one tooth. A patient may have multiple cases for different teeth or treatments.

The system maintains case statuses, files, design versions, and case history throughout the case lifecycle.

2.2 Main Product Functions

Dentigo provides the following main functions:

1. User authentication and role-based access control.

2. User account management.

3. Patient, tooth, and clinical information management.

4. Treatment Plan Management.

5. Dental Case creation and validation.

6. Radiology file and report management.

7. Dental laboratory workflow and design version management.

8. Dentist review, revision requests, and design approval.

9. Case completion and cancellation.

10. Case history and activity tracking.

11. Dashboards, search, due-date monitoring, and notifications.

2.3 User Roles

2.3.1 Doctor

The Doctor can:

• View registered patients and dental cases within the Doctor role.

• Create and update patient information.

• Record clinical findings.

• Create Dental Cases.

• Create and manage Treatment Plans.

• Mark cases as requiring radiology.

• Upload relevant clinical files.

• Review laboratory designs.

• Request revisions or reject designs.

• Approve submitted designs.

• Complete or cancel cases according to the defined workflow.

• View the case history for accessible cases.

• Receive relevant notifications.

2.3.2 Lab Technician

The Lab Technician can:

• View cases assigned or made available to the laboratory workflow.

• View the information required to prepare dental designs.

• View tooth information, measurements when provided, instructions, and permitted files.

• Update the laboratory work status.

• Upload new dental design versions.

• View previous design versions.

• Respond to revision requests.

• View permitted laboratory-related case history.

• Receive relevant notifications.

The Lab Technician cannot access radiology images or radiology reports.

2.3.3 Radiology User

The Radiology User can:

• View cases marked as requiring radiology in the authorized radiology queue.

• View the patient and tooth information needed to identify the case.

• Upload required radiology images.

• Upload the radiology report.

• View permitted radiology files and radiology-related case history.

• Receive relevant notifications.

The Radiology User cannot access unnecessary laboratory-only information or manage dental treatment decisions.

2.3.4 Secretary

The Secretary can:

• Register patients and update permitted contact details.

• Organize, confirm, reschedule, and cancel appointments according to clinic rules.

• Access a concise case-progress summary automatically generated by the system from existing permitted case information before the patient’s visit. The Secretary does not manually author clinical summaries.

• Record payments after they have been received at the clinic, including the amount, date, payment method, and payment status.

• View permitted patient, appointment, and payment information.

The Secretary cannot make clinical decisions, edit the doctor’s clinical findings or treatment decisions, review or approve dental designs, or change permissions.

2.3.5 Patient

The Patient is represented only as a patient record in the MVP and does not have a system account or login. Authorized clinic staff manage the patient's appointments and record payments received at the clinic. The patient cannot access Dentigo pages or records directly.

The Patient cannot log in to the system, access system pages directly, or view internal clinical notes, laboratory information, other patients’ records, or administrative functions.

2.3.6 Administrator

The Administrator can:

• Create user accounts.

• Edit basic account details.

• Change an account's role among the existing system roles.

• Activate or deactivate accounts.

• View system-wide case information and statuses.

• Monitor delays and overdue cases.

• View case history in read-only mode.

The Administrator cannot:

• Create Dental Cases.

• Edit clinical information.

• Approve designs.

• Request design revisions.

• Change Dental Case statuses.

• Cancel Dental Cases.

• Delete case history.

The Administrator manages existing roles only. The system does not support creating custom roles or defining custom permissions.

2.4 Operating Environment

The system is a web application accessible through a supported modern web browser.

The backend is responsible for business rules, authorization, data validation, file handling, and database operations.

2.5 General Constraints

• The system must enforce role-based permissions on the backend.

• Staff user accounts are created by the Administrator. Patients do not have system accounts; their information and appointments are managed by authorized clinic staff.

• User accounts are created by an Administrator.

• No AI functionality is included.

• In-app notifications are included; email and SMS notifications are excluded.

• Case history must be preserved.

• Previous file and design versions must not be overwritten.

• Physical delivery and pickup workflows are excluded from the MVP.

2.6 Assumptions

• Users have valid accounts created by an Administrator.

• Users have access to the internet and a supported browser.

• Doctors provide accurate patient, tooth, treatment, and clinical information.

• Required radiology files and reports are provided when a case needs radiology.

• Dental laboratory users upload designs in supported

formats.

• Each user is responsible for actions performed through their account.

3. External Interface and UI Requirements

3.1 General UI Requirements

The interface should:

• Display information according to the user's role.

• Provide clear navigation between the main sections.

• Display case statuses consistently.

• Show clear validation and error messages.

• Require confirmation before important actions when appropriate.

• Display loading, empty, success, and failure states.

• Provide clear feedback when a file upload succeeds or fails.

• Prevent users from accessing unauthorized pages, files, or actions.

The UI must reflect the actual permissions and workflow enforced by the backend.

3.2 Doctor Interface

The Doctor interface should include:

• Doctor Dashboard.

• Patient Management.

• Patient Details.

• Treatment Plan Management.

• Dental Case creation.

• Dental Case Details.

• Files and Versions.

• Case History.

• Doctor Review actions.

• Notifications.

The Doctor Dashboard may display:

• Active Cases.

• Doctor Review.

• Revision Requested.

• Completed Cases.

• Recent Dental Cases.

The following figures may be used as sample dashboard values:

• Active Cases: 24 (+12% this month).

• Doctor Review: 3.

• Revision Requested: 4.

• Completed Cases: 15 (All time).

These figures are illustrative UI values, not fixed system requirements.

Dashboard counting rules:

• Active Cases must exclude cases currently waiting for Doctor Review.

• Active Cases must exclude cases with Revision Requested status.

• Doctor Review and Revision Requested must be counted separately.

• Completed Cases may show the total number of completed cases over all time.

3.3 Lab Interface

The Lab interface should include:

• Lab Dashboard.

• Available Dental Cases.

• Case Details.

• Dental Design Upload.

• Design Versions.

• Revision Requests.

• Permitted Case History.

• Notifications.

The Lab Dashboard may include:

• Cases in progress.

• Designs submitted.

• Revision requests.

• Overdue cases.

The dashboard must not include a Materials section in the MVP.

The term Designs Submitted should be used instead of a general Completed label when the card refers only to

laboratory design submission.

Submitting a design does not mean the overall Dental Case is completed.

3.4 Radiology Interface

The system must provide a dedicated Radiology Dashboard.

The dashboard should display cases currently requiring radiology and available to the Radiology User.

The interface should provide:

• Radiology queue.

• Relevant case identification details.

• Radiology image upload.

• Radiology report upload.

• Upload status and validation messages.

• Permitted radiology-related case history.

A case must leave the active radiology queue when the required images and report have been successfully submitted.

Removing a case from the radiology queue does not mean the Dental Case is completed.

3.5 Administrator Interface

The Administrator interface should include:

• Administrator Dashboard.

• User Management.

• Account Details.

• Case Monitoring.

• Overdue Case Monitoring.

• Read-only Case Details.

• Read-only Case History.

The Administrator Dashboard must not contain a Create a Dental Case action.

3.6 File Upload Interface

The file upload interface must:

• Display supported file formats.

• Enforce the maximum file size.

• Display upload progress when supported.

• Display a clear error when an upload fails.

• Allow the user to retry a failed upload.

• Confirm successful uploads.

• Avoid registering incomplete or failed uploads as valid case files.

3.7 Search Interface

The system should support simple search using:

• Dental Case ID.

• Patient name.

Search results must be limited according to the user's role and permissions.

4. Functional Requirements

4.1 Authentication and Access Control

AUTH-01 — User Login

The system shall allow registered users to log in using their email address and password.

AUTH-02 — Account Creation

The system shall allow an Administrator to create user accounts.

AUTH-03 — No Self-Registration

The system shall not allow users to create their own accounts.

AUTH-04 — Role-Based Access

The system shall provide permissions according to the user's assigned role.

AUTH-05 — Backend Authorization

The backend shall validate permissions for protected operations and files. Hiding a button in the UI alone is not sufficient.

AUTH-06 — Deactivated Accounts

The system shall prevent deactivated users from logging in.

AUTH-07 — Logout

The system shall allow users to log out.

AUTH-08 — Session Expiration

The system shall support session expiration according to the implemented authentication mechanism.

AUTH-09 — Password Security

The system shall store passwords using an appropriate secure password-hashing mechanism.

4.2 User Management

USER-01 — Create Users

The Administrator shall be able to create accounts for Doctors, Secretaries, Lab Technicians, Radiology Users, and Administrators.

USER-02 — Edit User Details

The Administrator shall be able to edit basic account details.

USER-03 — Change Existing Roles

The Administrator shall be able to change a user's role to another existing role.

USER-04 — No Custom Roles

The system shall not support creating custom roles or customizing permissions in the MVP.

USER-05 — Account Activation

The Administrator shall be able to activate or deactivate accounts.

USER-06 — Preserve Account History

The system shall not permanently delete user accounts that may be referenced by case history or other records.

USER-07 — Protect the Last Active Administrator

The system shall prevent an operation that would leave the system without an active Administrator.

For example, the system must prevent deactivating or changing the role of the last active Administrator if no other active Administrator exists.

4.3 Patient, Tooth, and Clinical Information

PAT-01 — Patient Records

The system shall allow Doctors and authorized Secretaries to view registered patients according to their role permissions.

PAT-02 — Patient Information

The system shall allow the Secretary to register patients and update permitted basic contact information, including name and phone number. Doctors shall also be able to view and update patient information within their permissions. Clinical findings and treatment decisions remain restricted to Doctors.

PAT-03 — Shared Patient Visibility

Doctors shall be able to access registered patient information within the Doctor role permissions.

PAT-04 — Patient Deletion

The system shall prevent permanent deletion of a patient record when it is linked to Dental Cases or case history.

PAT-05 — Tooth Records

The system shall allow the relevant tooth to be identified and linked to a patient and Dental Case.

PAT-06 — One Tooth per Case

Each Dental Case shall be associated with exactly one patient and exactly one tooth.

PAT-07 — Multiple Treatments

A patient may have multiple Dental Cases for different teeth or treatments.

PAT-08 — Clinical Findings

The system shall allow Doctors to record clinical findings, such as a dental problem or examination note.

PAT-09 — Clinical Finding Relationships

A clinical finding shall be linked to a patient and tooth. It may also be linked to a Dental Case when relevant.

PAT-10 — Clinical Finding Updates

The system shall allow Doctors to update clinical findings. Important changes to clinical information shall be recorded with the responsible user and timestamp where applicable. Previous information must not be silently erased when its preservation is necessary for traceability.

4.4 Treatment Plan Management

TP-01 — Create Treatment Plans

The system shall allow Doctors to create Treatment Plans for organizing planned procedures.

TP-02 — Manage Procedures

A Treatment Plan may contain multiple procedures or treatment stages.

TP-03 — Link Procedures to Cases

Each procedure may be linked to a separate Dental Case.

TP-04 — Standalone Cases

The system shall allow Doctors to create a Dental Case without linking it to a Treatment Plan.

TP-05 — Simplified Scope

Treatment Plan Management shall focus on organizing procedures and linking them to Dental Cases.

Cost estimation, payments, invoices, and patient consent management are excluded from the MVP.

4.5 Dental Case Creation and Quality Gate

CASE-01 — Create Dental Case

The system shall allow Doctors to create a Dental Case.

CASE-02 — Required Case Information

A Dental Case shall include the required patient, tooth, treatment, and case information.

CASE-03 — Case Files

The Doctor shall be able to attach relevant files to a Dental Case.

CASE-04 — Radiology Requirement

The Doctor shall be able to mark a Dental Case as requiring radiology during creation or later in the workflow, subject to the defined permissions and status rules.

CASE-05 — Quality Gate

The system shall validate required case information and laboratory■required files before allowing submission to the laboratory.

CASE-06 — Conditional Radiology Validation

When radiology is required, the case shall enter the Radiology Queue until all required images and the report are submitted. Missing radiology files shall not, by themselves, prevent submission to the laboratory. Radiology completion and laboratory submission are tracked separately.

CASE-07 — Validation Feedback

The system shall display missing or invalid information when the Quality Gate fails.

CASE-08 — Case Submission

The Doctor may submit the case after required case information and laboratory files pass

the Quality Gate. Pending radiology files shall not block submission.

CASE-09 — Case Identification

The system shall assign a unique identifier to each Dental Case.

CASE-10 — Case Status

The system shall maintain the current status of each Dental

Case.

CASE-11 — Case Information Updates

The system shall enforce appropriate permissions and status rules when case information is updated.

Important changes must be traceable where applicable.

4.6 Radiology Workflow

RAD-01 — Radiology Queue

The system shall provide a queue of Dental Cases currently requiring radiology.

RAD-02 — Role-Based Queue Access

Radiology Users shall be able to view only cases available to their authorized radiology queue.

RAD-03 — Upload Radiology Images

The Radiology User shall be able to upload radiology images for an authorized case.

RAD-04 — Upload Radiology Report

The Radiology User shall be able to upload the required radiology report.

RAD-05 — Correct Record Linking

Radiology images and reports shall be linked to the correct patient, Dental Case, and tooth.

RAD-06 — Required Submission

Radiology is complete only when all required images and the report are successfully uploaded. This controls Radiology Queue membership; it does not block laboratory submission.

RAD-07 — Remove from Radiology Queue

When all required radiology files are uploaded, the case leaves the active Radiology Queue, even if laboratory work is ongoing.

RAD-08 — Preserve Main Case Status

Completing radiology uploads shall not automatically mark the Dental Case as Completed.

RAD-09 — Additional Radiology

A Doctor shall be able to mark a case as requiring additional radiology later in the workflow.

New radiology files shall be added without overwriting previously submitted files.

RAD-10 — Radiology File Privacy

The system shall prevent Lab Technicians from accessing radiology images and radiology reports.

RAD-11 — Radiology History

The system shall record relevant radiology actions in the case history.

A separate complex radiology state machine is not required for the MVP. The radiology queue is based on whether radiology is currently required and whether the required files have been submitted.

4.7 Laboratory Workflow and Design Versions

LAB-01 — View Available Cases

The Lab Technician shall be able to view cases available to the laboratory workflow.

LAB-02 — Access Required Information

The Lab Technician shall be able to view the information required to prepare a dental design.

This may include:

• Tooth name or number.

• Relevant measurements, when provided.

• Doctor's instructions.

• Files required for dental design.

LAB-03 — Restricted Radiology Access

The Lab Technician shall not be able to view radiology images or reports.

LAB-04 — Upload Design

The Lab Technician shall be able to upload a dental design for a Dental Case.

LAB-05 — Submit Design for Review

After uploading the design, the Lab Technician shall be able to submit it for Doctor Review.

LAB-06 — Design Versions

The system shall preserve each submitted design as a separate version.

LAB-07 — Preserve Previous Versions

Uploading a new design shall not overwrite previous versions.

LAB-08 — View Previous Versions

Authorized users shall be able to view previous design versions.

LAB-09 — Compare Versions

The system should allow authorized users to compare available design versions when the supported file types and interface allow meaningful comparison.

LAB-10 — Revision Requests

The Lab Technician shall be able to view revision requests and their recorded reasons.

LAB-11 — Resubmit Design

The Lab Technician shall be able to upload and submit a revised design after a revision request.

LAB-12 — Design Submission Is Not Completion

Submitting a design shall not automatically mark the overall Dental Case as Completed.

LAB-13 — Design Version Metadata

Each submitted design version shall record relevant metadata, including the uploader, upload time, and version identifier.

4.8 Doctor Review and Design Approval

REV-01 — Review Queue

The system shall provide Doctors with access to cases awaiting their review.

REV-02 — View Submitted Design

The Doctor shall be able to view the latest submitted design and relevant permitted case information.

REV-03 — Approve Design

The Doctor shall be able to approve the latest submitted design.

REV-04 — Request Revision

The Doctor shall be able to request a design revision.

REV-05 — Reject Design

The Doctor shall be able to reject an unsuitable design and return the case to the laboratory for rework.

REV-06 — Reason Required

The Doctor shall provide a reason when requesting a revision or rejecting a design.

REV-07 — Preserve Rejected Designs

Rejecting a design shall not delete the submitted design or its version history.

REV-08 — Record Review Actions

Approval, revision requests, and rejection actions shall be recorded in the case history with the responsible user and timestamp.

REV-09 — Latest Version Rule

Approval or revision decisions shall apply only to the latest submitted design version.

REV-10 — Prevent Outdated Decisions

The backend shall prevent a Doctor from approving or requesting a revision for an outdated design version when a newer version has already been submitted.

REV-11 — Approval Meaning

The Approved status shall mean that the Doctor has accepted the submitted design.

It shall not automatically mean that the overall treatment or Dental Case is completed.

4.9 Case Completion

REV-12 — Complete Case

The system shall allow an authorized Doctor to mark a Dental Case as Completed when the overall workflow has finished.

REV-13 — Completion Validation

The system shall validate the applicable completion conditions before allowing a case to be marked as Completed.

REV-14 — Preserve Completed Cases

Completed cases shall remain available for authorized viewing and history review.

REV-15 — Completion History

The system shall record the completion action, including the responsible user and timestamp.

4.10 Case Cancellation

CAN-01 — Cancellation Permission

Only a Doctor shall be allowed to cancel a Dental Case.

CAN-02 — Cancellation Reason

The Doctor shall provide a cancellation reason.

The available reasons shall include:

• Created by mistake.

• Wrong patient.

• Duplicate case.

• Treatment no longer required.

• Other.

CAN-03 — Other Reason

If the Doctor selects Other, the system shall require a written explanation.

CAN-04 — Cancel Cases at Different Stages

A Doctor may cancel a case after creation, including a case already submitted to the laboratory, subject to the workflow rules.

CAN-05 — Preserve Cancelled Cases

Cancellation shall not permanently delete the Dental Case, its files, or its history.

CAN-06 — Record Cancellation

The system shall record the cancellation reason, responsible user, and timestamp.

CAN-07 — Administrator Restriction

The Administrator shall not be allowed to cancel Dental Cases.

4.11 Case History

HIST-01 — Record Important Events

The system shall maintain a chronological history of important actions and changes related to each Dental Case.

HIST-02 — History Information

Where applicable, each history event shall include:

• Event type or description.

• Responsible user.

• Timestamp.

• Relevant status change.

• Related file or design version.

• Reason for revision, rejection, or cancellation when applicable.

HIST-03 — Doctor History Access

Doctors shall be able to view the full history of cases they are authorized to access.

HIST-04 — Lab History Access

Lab Technicians shall be able to view laboratory-related events and permitted case information.

They shall not be able to view radiology files or restricted radiology details.

HIST-05 — Radiology History Access

Radiology Users shall be able to view radiology-related events for cases available in their authorized queue.

They shall not be given unnecessary laboratory-only details.

HIST-06 — Administrator History Access

Administrators shall be able to view the full case history in read-only mode.

HIST-07 — Preserve History

The application shall not provide users with the ability to edit or delete existing case history events.

HIST-08 — Backend Enforcement

The backend shall enforce history visibility according to the user's role and permissions.

4.12 Dashboards, Search, and Due Dates

DASH-01 — Role-Specific Dashboards

The system shall provide a dashboard appropriate to each user role.

DASH-02 — Doctor Dashboard

The Doctor Dashboard shall display useful case counts and recent cases.

DASH-03 — Lab Dashboard

The Lab Dashboard shall display relevant laboratory work, submitted designs, revision requests, and overdue cases.

DASH-04 — Radiology Dashboard

The Radiology Dashboard shall display cases currently requiring radiology.

DASH-05 — Administrator Dashboard

The Administrator Dashboard shall support system-wide case monitoring without allowing clinical case modifications.

DASH-06 — Expected Due Date

A Dental Case may have an expected due date.

DASH-07 — Overdue Rule

A case shall be considered overdue when its expected due date has passed and the relevant work remains incomplete.

DASH-08 — No Separate Review Deadline

A separate due date for Doctor Review is not required in the MVP.

DASH-09 — Search by Case ID

The system shall allow users to search for Dental Cases by their identifiers, subject to role-based access.

DASH-10 — Search by Patient Name

The system shall allow authorized users to search using a patient's name.

DASH-11 — Search Results

Search results shall contain only records the user is permitted to access.

4.13 Notifications

NOT-01 — In-App Notifications

The system shall provide in-app notifications for relevant workflow events.

NOT-02 — Notification Events

Notifications may be generated for:

• A design awaiting Doctor Review.

• A revision being requested.

• A new design being submitted.

• Radiology being required.

• A case being completed.

• A case becoming overdue.

NOT-03 — Role-Appropriate Notifications

Notifications shall be sent only to users for whom the event is relevant.

NOT-04 — Link to Dental Case

A notification related to a Dental Case shall provide a way to open the corresponding case when the recipient has permission to access it.

NOT-05 — No Email or SMS

Email and SMS notifications are excluded from the MVP.

4.14 File Management

FILE-01 — Supported File Types

The system shall support the following file types:

• PDF.

• JPG.

• PNG.

• STL.

• OBJ.

FILE-02 — Maximum File Size

The system shall enforce a maximum file size of 25 MB per file.

FILE-03 — File Validation

The system shall validate file type and size before accepting an upload.

FILE-04 — Upload Failure

If an upload fails, the system shall display an understandable error and allow the user to retry.

FILE-05 — Successful Upload

A file shall be registered as a valid case file only after a successful upload.

FILE-06 — File Relationships

Files shall be linked to the appropriate patient and Dental Case. Tooth association shall be recorded where relevant.

FILE-07 — File Version Preservation

New design versions and additional radiology files shall not overwrite existing files.

FILE-08 — File Access Control

The backend shall validate the user's permissions before allowing file preview or download.

FILE-09 — File Metadata

The system shall maintain relevant file metadata, including file type, size, upload time, uploader, and associated case.

FILE-10 — File History

Important file uploads and version submissions shall be recorded in the case history.

4.15 Appointment Management

FR-APT-01: The system shall allow the Secretary to create and book appointments for registered Patients, subject to appointment availability and doctor scheduling rules.

FR-APT-02: The system shall allow authorized clinic staff to view and manage

appointment date, time, assigned Doctor, and appointment status. Patients shall

not access this information through a patient-facing account.

FR-APT-03: The Secretary shall be able to create, confirm, reschedule, and cancel

appointments according to clinic rules.

FR-APT-04: The system shall prevent conflicting bookings for the same Doctor and

time slot.

FR-APT-05: The Secretary and Doctor shall be able to view their permitted

appointment schedules.

FR-APT-06: The system shall maintain appointment status and relevant changes

for traceability.

FR-APT-07: Each appointment shall be associated with a Patient and an assigned Doctor. A Dental Case may be linked to an appointment when applicable.

FR-APT-08: The system shall automatically generate a concise case-progress summary from existing permitted case information for the Doctor before the Patient's appointment. The Secretary may view or access the generated summary but shall not be required to write a clinical summary manually.

4.16 Clinic Payment Records

FR-PAY-01: The system shall allow the Secretary or another specifically authorized staff user to record a payment only after the clinic has received it.

FR-PAY-02: Each payment record shall include the Patient, amount, payment date,

payment method, payment status, and the staff user who recorded it.

FR-PAY-03: A payment may be associated with a Dental Case or treatment when

applicable.

FR-PAY-04: The system shall allow authorized staff to view payment records and

payment status.

FR-PAY-05: The system shall allow authorized clinic staff to view payment records.

Patients shall not access payment records directly through Dentigo.

FR-PAY-06: The system shall preserve payment record history and shall not treat

an appointment booking as proof of payment.

FR-PAY-07: The system shall not collect or process online payments. Payment takes place at the clinic, and Dentigo records the payment only after it has been received.

FR-PAY-08: Any correction or cancellation of a recorded payment shall be

restricted to authorized staff and recorded for traceability.

5. Business Rules and Workflows

5.1 Main Dental Case Workflow

The main case workflow is:

1. The Doctor selects a patient.

2. The Doctor selects one tooth.

3. The Doctor selects a treatment.

4. The Doctor enters the required case information.

5. The Doctor attaches relevant files.

6. If radiology is required, the case enters the radiology

7. If radiology is required, the case enters the Radiology Queue. Radiology

completion is tracked separately from laboratory submission.

8. The Quality Gate validates the required case information and files needed

for

laboratory work; missing radiology files alone do not cause it to fail.

9. After the Quality Gate passes, the Doctor may submit the case to the

laboratory even

if radiology uploads are still pending.

10. The Lab Technician processes the case and uploads a

dental design.

11. The Lab Technician submits the design for Doctor

Review.

12. The Doctor reviews the latest submitted design.

13. The Doctor approves the design or requests a

revision.

14. If revision is requested, the case returns to the

laboratory for rework.

15. The process continues until the required design is

approved and the overall case workflow is finished.

16. The Doctor marks the case as Completed when

applicable.

Cancellation may occur at an allowed stage and requires a

reason.

5.2 Case Status Definitions

The system shall use the following statuses:

Status	Meaning
Draft	The case has been created but has not been submitted to the laboratory.
Submitted	The case has been submitted and is awaiting laboratory processing.
In Progress	The laboratory is working on the case.
Design Submitted / Doctor Review	A design has been submitted and is waiting for the Doctor's decision.
Revision Requested	The Doctor has requested changes to the submitted design.
Approved	The Doctor has accepted the latest submitted design.
Completed	The overall Dental Case workflow has finished.
Cancelled	The Doctor has cancelled the case with a recorded reason.

The system shall not treat Approved and Completed as equivalent statuses.

Radiology queue membership is managed separately from the main Dental Case status.

5.3 Design Revision Workflow

When the Doctor requests a revision or rejects a design:

1. The Doctor must provide a reason.

2. The action is recorded in Case History.

3. The case returns to the laboratory workflow.

4. The Lab Technician can review the reason.

5. The Lab Technician uploads a new design version.

6. The previous version remains available.

7. The new version is submitted for Doctor Review.

5.4 Radiology Queue Rules

A case enters the active Radiology Queue when radiology is required and files are missing. It remains there whether or not the case has been submitted to the laboratory.

• The Radiology User can access only authorized cases in that queue.

• The case leaves the active queue when all required images and the report have been submitted successfully.

• Completing radiology uploads does not complete the Dental Case.

• If the Doctor later requires additional radiology, the case can re-enter the queue.

• New radiology files must not replace previously submitted files.

5.5 File Version Rules

• Each submitted dental design must be stored as a distinct version.

• A new design version must not overwrite an earlier version.

• Previous versions must remain available to authorized users.

• A review decision must apply to the latest submitted design version.

• The backend must reject decisions based on an outdated version when a newer version exists.

5.6 Overdue Case Rules

A case is overdue when:

1. An expected due date exists.

2. The due date has passed.

3. The relevant work is still incomplete.

The system does not require a separate deadline for Doctor Review in the MVP.

5.7 Case Cancellation Rules

• Only Doctors can cancel Dental Cases.

• A cancellation reason is mandatory.

• Selecting Other requires a written explanation.

• Cancellation does not delete files, case history, or the case itself.

• Administrators cannot cancel cases.

6. Non-Functional Requirements

6.1 Security

NFR-SEC-01

The system shall authenticate users before allowing access to protected functions.

NFR-SEC-02

The backend shall enforce authorization for all protected operations.

NFR-SEC-03

The system shall protect patient information and restrict access to sensitive files.

NFR-SEC-04

Passwords shall be stored using secure password hashing.

NFR-SEC-05

File access shall be authorized on the server side.

NFR-SEC-06

The system shall validate user input and uploaded files.

6.2 Data Integrity

NFR-DATA-01

Each Dental Case shall be linked to one patient and one tooth.

NFR-DATA-02

The system shall preserve important case history events.

NFR-DATA-03

The system shall preserve previous design versions and radiology files.

NFR-DATA-04

Important operations shall not leave records in an inconsistent state.

NFR-DATA-05

The system shall prevent unauthorized or invalid workflow transitions.

6.3 Reliability

NFR-REL-01

The system shall provide clear feedback when an operation fails.

NFR-REL-02

A failed file upload shall not be treated as a successful upload.

NFR-REL-03

Important case actions shall be saved consistently.

6.4 Usability

NFR-USE-01

The interface shall provide clear navigation and understandable labels.

NFR-USE-02

The interface shall display statuses consistently across relevant screens.

NFR-USE-03

The system shall explain validation errors and missing required information.

NFR-USE-04

The interface shall display actions according to the user's permissions.

6.5 Maintainability

NFR-MAINT-01

The backend should be organized into understandable components.

NFR-MAINT-02

Business rules should be implemented and validated on the backend rather than relying only on UI behavior.

NFR-MAINT-03

The system should use clear relationships between users, patients, teeth, cases, files, and history records.

6.6 Testability

NFR-TEST-01

The system shall allow testing of authentication and role permissions.

NFR-TEST-02

The system shall allow testing of case status transitions.

NFR-TEST-03

The system shall allow testing of the Quality Gate and independent radiology queue tracking.

NFR-TEST-04

The system shall allow testing of file upload restrictions and version preservation.

NFR-TEST-05

The system shall allow testing of cancellation and design review rules.

Specific performance or availability targets are not defined in this version and should not be assumed without stakeholder approval.

7. Data Requirements

The following entities represent the main data concepts of Dentigo. They describe the required information and relationships, not the final database schema.

7.1 User

Represents a person with access to the system.

Main information:

• User ID.

• Name.

• Email.

• Password hash.

• Role.

• Account status.

• Created date.

Supported roles:

• Doctor.

• Lab Technician.

• Radiology User.

• Administrator.

7.2 Patient

Represents a patient receiving dental treatment.

Main information:

• Patient ID.

• Name.

• Phone number.

• Relevant basic patient details.

• Created date.

• Updated date, when applicable.

A patient may have multiple teeth and multiple Dental Cases.

7.3 Tooth

Represents a tooth associated with a patient.

Main information:

• Tooth ID.

• Patient ID.

• Tooth identifier or number.

• Relevant tooth information.

A tooth belongs to a patient and may be associated with

multiple clinical findings and Dental Cases over time.

7.4 Clinical Finding

Represents a clinical observation or finding.

Main information:

• Finding ID.

• Patient ID.

• Tooth ID.

• Optional Dental Case ID.

• Description.

• Created by.

• Created date.

• Updated date, when applicable.

Important updates must be traceable.

7.5 Treatment Plan

Represents an organized plan for one or more dental procedures.

Main information:

• Treatment Plan ID.

• Patient ID.

• Plan name or description.

• Created by.

• Created date.

• Status, if required by the implementation.

A Treatment Plan may contain multiple procedures.

7.6 Procedure

Represents a treatment procedure or stage within a Treatment Plan.

Main information:

• Procedure ID.

• Treatment Plan ID.

• Procedure name or description.

• Related Dental Case ID, when linked.

A procedure may be linked to a separate Dental Case. A Dental Case may also exist without a Treatment Plan.

7.7 Dental Case

Represents one treatment or procedure for one tooth of one patient.

Main information:

• Dental Case ID.

• Patient ID.

• Tooth ID.

• Treatment or procedure.

• Case description and instructions.

• Current status.

• Radiology-required flag.

• Expected due date, when applicable.

• Created by.

• Created date.

• Updated date, when applicable.

• Cancellation reason, when applicable.

7.8 Case File

Represents a file associated with a patient and Dental Case.

Main information:

• File ID.

• Patient ID.

• Dental Case ID.

• Tooth ID, when relevant.

• File name.

• File type.

• File size.

• Storage reference.

• Uploaded by.

• Upload date.

• File category.

• Version information, when applicable.

Supported formats are PDF, JPG, PNG, STL, and OBJ. Maximum file size is 25 MB per file.

7.9 Design Version

Represents a submitted dental design version.

Main information:

• Version ID.

• Dental Case ID.

• Version number or identifier.

• File reference.

• Submitted by.

• Submission date.

• Review result, when applicable.

• Related revision reason, when applicable.

Each version must remain distinguishable from earlier versions.

7.10 Case History Event

Represents an important action or change in the Dental Case workflow.

Main information:

• Event ID.

• Dental Case ID.

• Event type.

• Description.

• Actor ID, when applicable.

• Timestamp.

• Previous status, when applicable.

• New status, when applicable.

• Related file or version, when applicable.

• Reason, when applicable.

History events must be preserved and displayed according to role permissions.

7.11 Notification

Represents an in-app notification.

Main information:

• Notification ID.

• Recipient user ID.

• Related Dental Case ID, when applicable.

• Notification type.

• Message.

• Created date.

• Read status.

Notifications must be restricted to authorized recipients.

7.12 Appointment

The Appointment entity represents a scheduled visit and should include an identifier, PatientId, DoctorId, appointment date and time, status, optional DentalCaseId, creation timestamp, and any required rescheduling or cancellation details. The appointment must be related to a Patient and a Doctor. The final database design should define the allowed status values and prevent conflicting bookings for a Doctor.

7.13 Payment

The Payment entity represents a payment received by the clinic and recorded in Dentigo. It should include an identifier, PatientId, optional DentalCaseId, amount, payment date, payment method, payment status, RecordedByUserId, and creation timestamp. Amounts must use a decimal data type. Payment records must not imply online payment processing. The final database design should define allowed payment methods and statuses and preserve auditability.

8. MVP Scope and Exclusions

8.1 Included in the MVP

The MVP includes:

1. Authentication and role-based permissions.

2. User management for the supported staff roles, including Doctor, Secretary,

Lab Technician, Radiology User, and Administrator.

3. Patient, tooth, and basic clinical information management.

4. Simplified Treatment Plan Management.

5. Dental Case creation and Quality Gate validation.

6. Conditional radiology workflow.

7. Radiology image and report uploads.

8. Laboratory workflow and dental design uploads.

9. Design version preservation.

10. Doctor review, revision requests, rejection, and

approval.

11. Case completion and cancellation.

12. Role-based Case History.

13. Dashboards and basic case search.

14. Expected due dates and overdue monitoring.

15. In-app notifications.

16. File validation and access control.

17. Secretary-managed appointment scheduling and appointment status tracking.

18. Clinic payment records entered by authorized staff after payment is received.

19. Automatic case-progress summaries for Doctors, generated from existing permitted case data.

8.2 Excluded from the MVP

The following features are excluded:

- AI-based diagnosis or treatment recommendations.
- AI image analysis.
- Patient self-registration.
- Custom role creation or custom permission management.

• Online payment collection or payment-gateway integration.

• Invoices and broader financial/accounting management.

• Treatment cost estimation.

• Patient consent management.

• Email and SMS notifications.

• Physical delivery and pickup tracking.

• Permanent deletion of cases and their histories.

• Complex radiology request state machines.

• Advanced analytics not required for the main workflow.

Any additional feature should be reviewed against the project's timeline and implementation capacity before being added to the MVP.

Appendix A — Role Permission Matrix

Lab

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permissions

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Edit basic

Yes No No No

patient details

Create Dental

Yes No No No

Case

Lab

Radiolog Administra

Function Doctor Technicia

y User tor

n

Create

Yes No No No

Treatment Plan

Mark case as

requiring Yes No No No

radiology

Read-only

Yes, for access

View radiology authorize according

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Complete case Yes No No No
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Cancel case No No No
reason
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View full Case Yes, read-
accessible No No
History only
cases
View Yes, for No,
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Yes, for Yes, for
View radiology- Yes, read-
accessible No authorize
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No No No Yes

accounts

Change existing

No No No Yes

user roles

Activate/deactiv

No No No Yes

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Lab

Radiolog Administra

Function Doctor Technicia

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Change Dental d

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Case status laborator No No

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Note: The backend must enforce this matrix. The frontend

should not be the only security control.

Appendix B — Status and Notification Reference

B.1 Main Case Statuses

• Draft.

• Submitted.

• In Progress.

• Design Submitted / Doctor Review.

• Revision Requested.

• Approved.

• Completed.

• Cancelled.

Radiology queue status is managed separately.

B.2 Main Notification Events

Event Intended Recipient

A design is submitted for

Doctor

review

A revision is requested Lab Technician

A new design version is

Doctor

submitted

Radiology is required Authorized Radiology User

Radiology files are

Relevant Doctor

submitted

Relevant users according to

A case is completed

workflow

Relevant users according to

A case becomes overdue

monitoring permissions

Appendix C — UI Design Consistency Checklist

Use this checklist when reviewing the Figma screens against the

SRS.

Doctor Dashboard

• Active Cases excludes Doctor Review and Revision

Requested.

• Doctor Review and Revision Requested appear as

separate counts.

• Dashboard figures are treated as sample data.

• Recent Dental Cases show useful case information.

Lab Dashboard

• Remove the Materials section.

• Use Designs Submitted instead of Completed for

laboratory design submission.

• Keep Overdue based on the expected due date and

incomplete work.

• Make revision requests and their reasons visible.

Radiology Dashboard

• Add a dedicated Radiology Dashboard.

• Show only cases currently requiring radiology and available to the Radiology User.

• Provide image and report upload.

• Show upload success and failure states.

• Remove the case from the active queue when all required files are submitted.

• Do not mark the overall Dental Case as Completed after radiology submission.

Administrator Dashboard

• Remove Create a Dental Case.

• Keep case information read-only.

• Keep case history read-only.

• Add Radiology User to the User Management role list.

• Include account activation and deactivation.

• Do not provide clinical case actions.

Create Dental Case

• Include Patient, Tooth, Treatment, Case Information, Files, Review, and Quality Gate.

• Require exactly one patient and one tooth per case.

• Allow standalone cases without a Treatment Plan.

·• Radiology files are required to complete radiology, not to submit the case to the laboratory.

• Display missing information when the Quality Gate fails.

Case Details

• Display the current case status clearly.

• Display case timeline and history.

• Display Files and Versions.

• Preserve previous design versions.

• Keep Approve Design and Reject Design as distinct actions.

• Require a reason for revision or rejection.

• Prevent decisions on outdated design versions.

• Distinguish Approved from Completed.

• Provide cancellation with a mandatory reason for

Doctors only.

Case History

• Display events chronologically.

• Include the responsible user and timestamp when applicable.

• Show only the events and files allowed for each role.

• Prevent editing or deleting history through the application.

Files

• Display supported formats: PDF, JPG, PNG, STL, and OBJ.

• Enforce the 25 MB maximum per file.

• Display upload errors and allow retries.

• Do not overwrite previous versions.

• Restrict file preview and download according to permissions.

End of Document

Version Note: This consolidated SRS defines the agreed working requirements for Dentigo v1.2 and is the reference for Figma alignment, backend design, database design, and MVP implementation. Scope or workflow changes should be reviewed and documented before implementation.